MILITARY SPECIFICATION

NAVIGATIONAL SET, AN/URN-25, TACAN

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for production of a Tactical Air Navigation (TACAN) all-band transponder, Navigational Set, AN/URN-25, hereinafter referred to as the navigational set, for use in conjunction with a TACAN antenna group to provide navigational information to aircraft.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbook, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging Of
MIL-M-38510	Microcircuits, General Specification For

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-740	Airborne And Structureborne Noise Measurements And Acceptance Criteria Of Shipboard Equipment
MIL-STD-781	Reliability Design Qualification And Production Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
MIL-STD-1310	Shipboard Bonding, Grounding, And Other Techniques For Electromagnetic Compatibility
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
MIL-STD-1387	Procedures For Submission Of Application For Approval Of Non-Standard General Purpose Electronic Test Equipment
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-003-121), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5825

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MS3114 — Connector, Receptacle, Electric, Series 1, Solder Type, Jam Nut Mounting, Bayonet Coupling

HANDBOOK

MILITARY

MIL-HDBK-241 — Design Guide For Electromagnetic Interference (EMI) Reduction In Power Supplies

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

NAVAL MILITARY PERSONNEL COMMAND (NMPC)

NAVPERS 18068 — Navy Enlisted Manpower And Personnel Classifications And Occupational Standards

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536E with SCN 1 — Process Specification Procedures And Requirements For Preparation And Soldering Of Electrical Connections

(Copies of specifications, standards, handbooks, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-310-C-77 — Racks, Panels And Associated Equipment

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The navigational set shall be in accordance with MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Navigational set description. The navigational set shall consist of a beacon cabinet and a control-indicator unit (see 6.4.4) suitable for ship or shore installation. The navigational set shall provide 100 properly equipped aircraft with range, bearing, and identification information within a 321.8 kilometer (km) (200 mile (mi)) line-of-sight range. The navigational set shall have a 252 channel capability and require a maximum of 3 minutes to change from any channel to any other channel without external test equipment or internal adjustments. Vacuum tubes shall not be used except for the last two radio frequency (RF) power stages.

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3.2.1 Major components. The beacon shall consist of the major components specified in a and b, enclosed in one cabinet:

a. Coder-keyer
b. Receiver-transmitter

3.2.1.1 Major functions. The functions specified in a through d shall be contained in separate drawers or contained within the major components of the beacon cabinet:

a. Monitoring circuitry (see 3.3.3)
b. Control-interface circuitry
c. Blowers (if required)
d. Secondary power supplies

3.2.1.2 Control-indicator. The control-indicator shall provide remote control operation and report beacon status.

3.3 Specific performance requirements. Specific performance requirements shall be as specified in 3.3.1 through 3.3.1.2.3.

3.3.1 Coder-keyer. The coder-keyer shall provide the composite video pulse train to the receiver-transmitter.

3.3.1.1 Coder. The coder shall develop the main and auxiliary reference groups, identity signal, establish priority between reference groups, identity, random, and distance reply pulses and provide pulse pair coding.

3.3.1.1.1 Main reference pulse group. The 15-hertz (Hz) reference group is called the main reference group. For X-channels, the main reference group shall consist of a group of 12 pairs of pulses, and the spacing between pairs shall be 30 ±0.3 microseconds (μs). For Y-channels, the main reference group shall consist of a group of 13 single pulses spaced 30 ±0.3 μs apart.

3.3.1.1.2 Auxiliary reference pulse group. The 135-Hz reference group is called the auxiliary reference group. For X-channels, the auxiliary reference group shall consist of six pairs of pulses, and the spacing between pairs shall be 24 ±0.25 μs. For Y-channels, the auxiliary reference group shall consist of a group of 13 single pulses spaced 15 ±0.2 μs apart.

3.3.1.1.3 Precedence. Reference groups shall have priority over all other portions of the output signal. Identity pulses shall have priority over all but reference groups. Distance reply pulses shall have priority over random pulses only.

3.3.1.1.4 Pulse coding. For X-channels, the pulses shall be coded in pairs with a spacing which shall be 12 ±0.25 μs. For Y-channels, the spacing shall be 30 ±0.25 μs. Delay compensation in the X-channel timing will be accomplished in the antenna group.

3.3.1.1.5 Antenna timing triggers. The main and auxiliary timing triggers provided to the beacon from the antenna group shall properly operate the burst development circuits. Their characteristics are: Amplitude of +7.0 ±1.5 volts (V) and -7.0 ±1.5 V measured across 10 kilohms shunted by 0.005 microfarad. These loads shall be incorporated into the transponder. The trigger's positive excursion occurs first and each half alternation has a length of 115 ±10 μs. The decay time of the positive to negative crossover does not exceed 2 μs. The leading and trailing edges of the rise and fall of the pulse are approximately 30 μs as shown in FIGURE 1.

3.3.1.1.6 Pretrigger. A pretrigger output from the beacon shall be provided. The pretrigger output shall be available at a receptacle located on the cable entrance box.

3.3.1.1.6.1 Pretrigger characteristics. The pretrigger shall have a lead time of 3.0 μs to 10 μs, measured to the 10 percent point on the rise of the transmitter pulse pair, with a stability of ±0.25 μs. The pretrigger amplitude shall be between +5 V and +50 V, with a pulse-width between 2.0 μs and 20 μs. The output impedance shall be 125 ohms. The rise time shall be less than 100 nanoseconds as measured at the end of 30.48 meter (m) (100 feet (ft)) of RG-58/U coaxial cable.

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3.3.1.2 Keyer. A solid-state electronic keyer capable of keying the transmitter by setting three lettered switches, one numbered switch, and a control switch to select normal, continuous or no tone, shall be provided. The lettered and numbered switch modules shall be capable of being by-passed and the last digit of the keyer control shall be numerical. In the event of antenna shutdown, the 1350-Hz oscillator shall continue to operate within the frequency tolerance of the phase-locked operation. The keyer controls shall be inside the appropriate drawer.

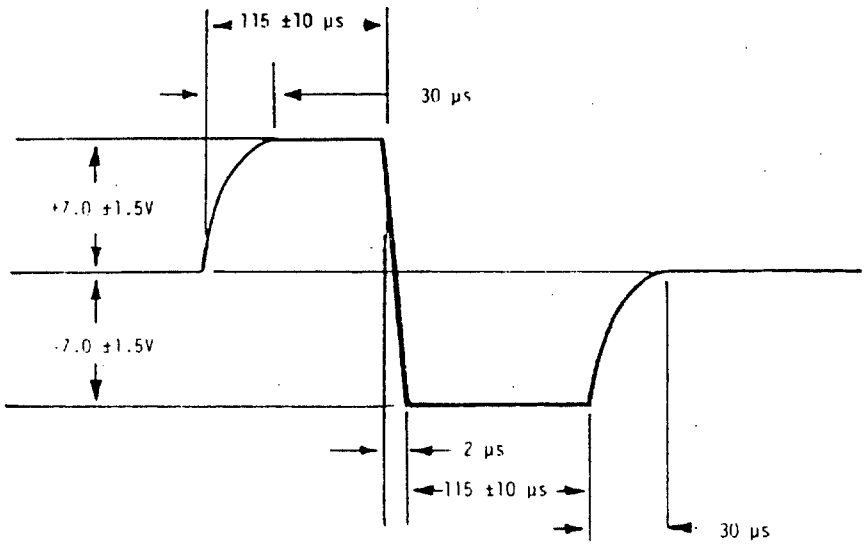


FIGURE 1. Trigger pulse shape.

3.3.1.2.1 Identification signal. The identification signal shall consist of a series of paired pulses spaced as specified in 3.3.1.1.4, transmitted at a repetition rate of 1350 pulse pairs per second (s) and phase-locked to the 135-Hz antenna trigger so that the first pulse of an identity pair shall occur 740 ±50 µs after the first pulse of the preceeding reference group.

3.3.1.2.2 Equalizing pulse pair. An equalizing pair of pulses shall be transmitted 100 ±10 µs after each identity pair.

3.3.1.2.3 Identification cycle. The random and distance reply pulses shall be replaced by the identity pulses at a recurrence rate of once every 30 s ±10 percent. During identity, the beacon code shall be transmitted as dots and dashes (Morse) of identity pulses, the spaces between dots and dashes being occupied by distance reply and random pulses. The dots and spaces between dots and dashes shall be between 100 milliseconds (ms) and 125 ms. Dashes and spaces between alphanumeric characters shall be between 300 ms and 375 ms.

3.3.2 Receiver-transmitter. The receiver shall contain, as a minimum, the pre-selector, mixer, RF amplifiers, detector, and first video amplifier. The transmitter shall contain, as a minimum, the RF frequency source, RF amplifiers, associated modulators, and monitoring and control circuitry. Low, medium, and high power levels shall be available as specified herein.

3.3.2.1 Receiver. The receiver shall be of a sensitive superheterodyne type, a passive transmit-receive switch, a decoder designed to process distance interrogation pulses, and a noise generator to maintain a constant output during conditions of less than full interrogation load. The receiver shall use a continuous wave (CW) output from the transmitter frequency source for local oscillator power. A YIG type pre-selection controlled by the transmitter channel select switches shall be used. A single log intermediate frequency (IF) is desired, as is single IF conversion. Receiver performance requirements shall be considered as beacon requirements unless otherwise specified herein.

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3.3.2.1.1 Frequency. The receiver shall be capable of being tuned to any one of the 126 1-megahertz (MHz) channels centered on integral frequencies between 1025 MHz and 1150 MHz.

3.3.2.1.2 Triggering level. The receiver shall respond to properly coded interrogations with a response percentage of at least 60 percent with a minimum signal strength of -90 decibels referred to one milliwatt (dBm) under conditions of 90 percent of full interrogation load. For a no-load condition, the minimum signal strength shall be -92 dBm with an interrogation pulse repetition frequency of no more than 150 pulse-pairs per s. The total number of pulses out of the receiver shall not vary from the normal operating setting by more than 3.3 percent for conditions of normal loading or for a 50 percent overload.

3.3.2.1.3 Demand mode. The beacon shall be provided with a switch to select Normal or Demand TACAN mode. In the Demand mode, the transmitter shall remain in a standby condition until an aircraft interrogates the beacon with a proper interrogation.

3.3.2.1.3.1 Triggering level. Under the no-load condition, when an interrogation is required to activate the transmitter, this activation shall occur when the minimum input signal strength is at most 3 decibels (dB) greater than the no-load sensitivity as specified in 3.3.2.1.2. A discernable interrogation period of 3 s to 4 s shall elapse before the transmitter turns on.

3.3.2.1.3.2 Delay time. When the interrogation rate drops to less than an average of 12 per s, an adjustable delay shall hold the transmitter in the on-air condition for a period of 1 minute to 3 minutes before returning to the standby condition.

3.3.2.1.4 Bandwidth. The bandwidth of the receiver shall be such that the triggering level shall not deteriorate by more than 3 dB when its total frequency drift is added directly to an incoming frequency drift of ±100 kilohertz (kHz).

3.3.2.1.5 Selectivity. The receiver shall reject properly coded interrogations centered on frequencies ±900 kHz removed from the nominal input frequency which have an amplitude of 80 dB above the receiver threshold. Interrogations arriving at all IFs shall be suppressed by 80 dB. Other spurious responses within 960 MHz to 1215 MHz and image frequencies shall be suppressed at least 65 dB.

3.3.2.1.6 Decoding. The receiver decoder timing shall be derived from the first pulse of an interrogation pair. Under conditions of interrogations by pulse pairs having a deviation from the nominal 12 µs or 36 µs by ±0.5 µs, the receiver triggering level shall not decrease by more than 1 dB. Under conditions when the pulse pair spacing deviates from nominal by more than ±2 µs, the receiver sensitivity shall be reduced by at least 70 dB. The receiver shall not respond to single pulse interrogations.

3.3.2.1.7 Response delay. The response delay, measured from the half amplitude point on the leading edge of the first interrogation pulse to the half amplitude point on the leading edge of the first reply pulse shall be 50 µs for X-channels and 56 µs for Y-channels. Tolerances shall be:

Input signal	Delay variation
0 dBm to -10 dBm	±0.5 µs
-10 dBm to -70 dBm	±0.25 µs
-70 dBm to -90 dBm	±0.8 µs

3.3.2.1.8 Dead time. The transponder dead time shall be measured from the trailing edge of a decoded pulse pair (from an interrogation or noise). The blanking shall be set at 60 ±1 µs for X-channels and 66 ±1 µs for Y-channels. The transponder shall be adjustable from these values by ±20 µs.

3.3.2.1.9 Echo suppression. The receiver shall contain an interrogation level sensing circuit to detect any interrogating pulse pair that is adjustable from 10 dB to 40 dB above threshold. The output of this level sensor shall be used to initiate receiver desensitization to any incoming signal level which is less than the preset threshold. The desensitizing gate shall be continuously adjustable between the limits of 150 µs to 400 µs. Any interrogation signal greater than the preset threshold shall be allowed to pass and shall initiate the desensitization period again regardless of whether it occurs during a previous desensitization period or not.

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3.3.2.1.10 Pulse count output. The total receiver pulse output from either interrogations or noise shall be controllable such that the total count can be varied from 2700 pulses per s to 4000 pulses per s.

3.3.2.1.11 CW interference. In the presence of on-channel and off-channel CW interference, the receiver shall respond to valid interrogations that exceed the CW interference by 10 dB or more. Conformance to this requirement shall be achieved with CW input signal strengths from threshold level to -30 dBm. The response delay tolerances, as a function of signal to CW interference, shall not exceed:

Level of signal level above CW level	Delay tolerances
10 dB to 20 dB	±1.2 μs
20 dB to 90 dB:	
Input signals equal to or less than -10 dBm	±0.375 μs
Input signals greater than -10 dBm	±0.75 μs

3.3.2.1.12 Narrow pulse interference. The receiver shall not reduce the traffic handling capability of the beacon in the presence of single input pulses or of pairs having the spacing within the tolerances of the decoder which have a duration of 1.2 μs or less.

3.3.2.1.13 Additional decoder capability. As a growth feature, adequate space shall be available in the coder-keyer card deck to add at least three additional circuit boards for future use.

3.3.2.1.14 Squitter distribution. Squitter distribution shall be as shown in FIGURE 2.

3.3.2.2. Transmitter. The transmitter shall be as specified in 3.3.2.2.1 through 3.3.2.2.7.6.

3.3.2.2.1 Frequency source. The frequency source shall be synthesizer controlled by channel numbered switches and shall provide 252 channels from 962 MHz through 1213 MHz. An illegal address (incorrect channel number), shall result in no output, and shall activate the illegal channel alarm. A direct phase-locked type is recommended. Spurious signal off of center frequencies shall be down greater than 95 dB at frequencies ±63 MHz from the operating frequencies and at least 70 dB down at the other frequencies in the band. The frequency source shall be directly interchangeable both physically and electrically with the monitor synthesizer.

3.3.2.2.1.1 Frequency lock time. The synthesizer shall be capable of locking on to any frequency, having been on any other frequency, in a maximum of 500 ms.

3.3.2.2.1.2 Frequency outputs. Two RF outputs shall be provided, one output to the first transmitter amplifier and the second output to the beacon receiver mixer.

3.3.2.2.1.3 Antenna control logic. The channel select circuitry shall provide, at the transponder cable entrance box, three separate two-line logic signals, derived from the channel setting, where logic 0 is -0.5 volts direct current (VDC) to +0.5 VDC, 20 milliamperes (mA) sinking current maximum. Logic 1 is +2.5 VDC to +5.0 VDC. Logic 1 will be a minimum of +3 VDC with 1 mA source current. The switching logic shall be as specified in TABLE I.

TABLE I. Switching logic.

Frequency (MHz)	Y mode	Band code
962 to 993	0	00
994 to 1024	0	01
1025 to 1055	1	00
1056 to 1087	1	01
1088 to 1119	1	10
1120 to 1150	1	11
1151 to 1181	0	10
1182 to 1213	0	11

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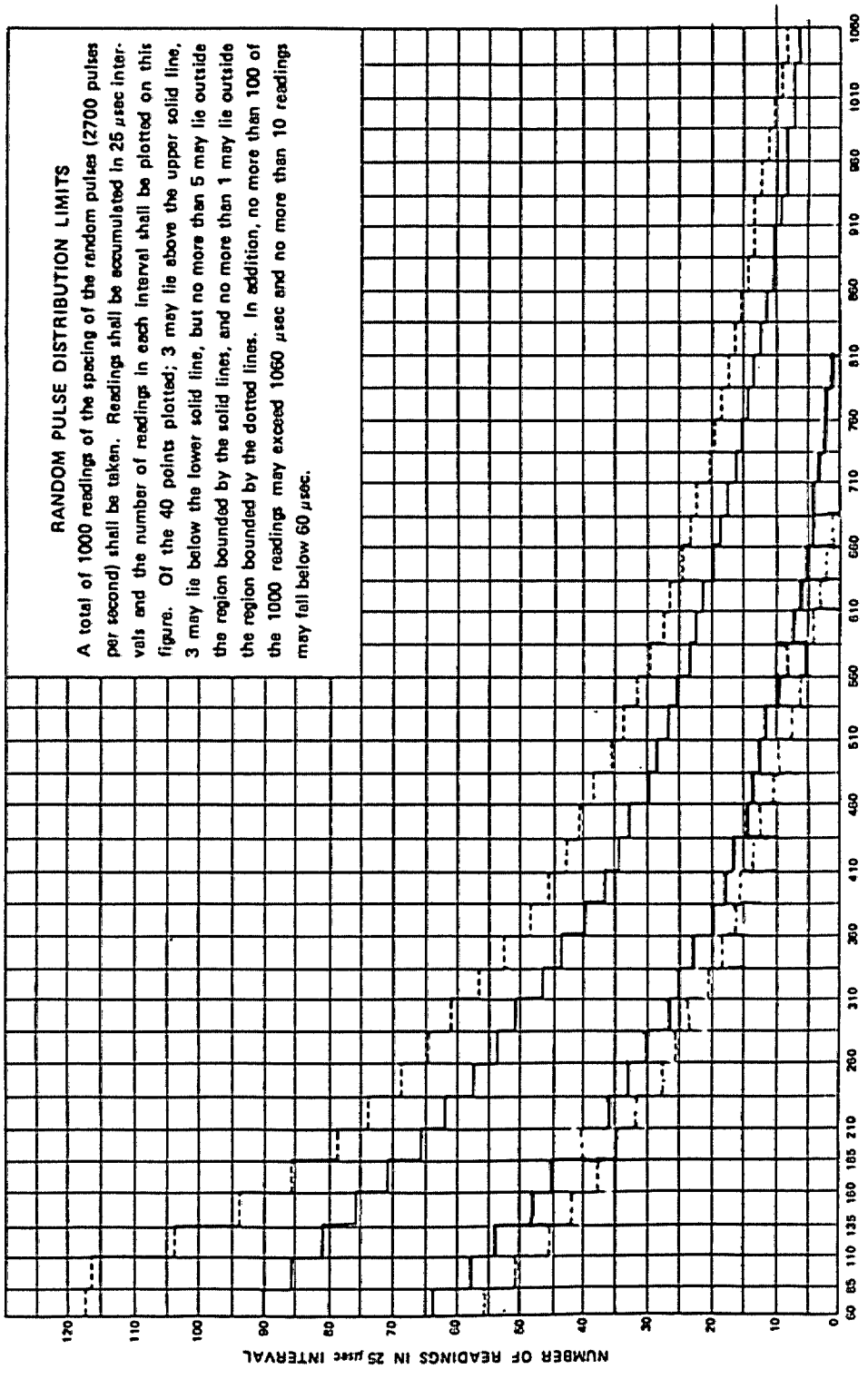


FIGURE 2. Pulse spacing in microseconds.

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3.3.2.2.1.4 YIG pre-selection control logic. The channel select circuitry shall provide the required logic to control the YIG pre-selector through the 1025 MHz to 1150 MHz tuning range.

3.3.2.2.2 Solid-state RF amplifier (driver). The solid-state amplifier shall amplify the input RF signal from the synthesizer, and, with its associated low level modulator, shall provide a suitable output to either the following RF cavity amplifier or as a Gaussian source for the first or low level beacon output (see 3.3.2.2.6). The solid-state amplifier shall be contained in a single module (exclusive of modulator) and shall be designed to operate without forced air cooling throughout the frequency, power, and temperature ranges. The solid-state amplifier shall not require user-operator required frequency tuning adjustments. The last RF output stage shall contain no more than two active devices. The amplifier shall be capable of operating into an open or shorted line without damage.

3.3.2.2.3 Low level amplifier modulator. The low level modulator shall provide the keying sequence and necessary pulse shaping to the solid-state amplifier. Adjustments shall not be necessary to maintain spectrum when changing from one channel to another. Preventive or corrective maintenance adjustments shall be kept to a minimum and be made with the aid of a general purpose multimeter and oscilloscope only.

3.3.2.2.4 Intermediate and final power amplifiers. The intermediate and final output shall be tube stages which are physically separate, identical, and interchangeable assemblies. These tube stages may share a common modulator and voltage source. The tubes shall be of the ceramic-metal planar triode type. Tube replacement shall not require tuning adjustments, other than front panel tuning, and no solder connections. Tubes procurable under the contractor's source control drawing shall conform to the minimum requirements of this specification. Total time to replace a tube shall be not more than 10 minutes, with no special tools required. Tuning controls shall be limited to not more than two per stage. The tuning controls shall have calibration. If calibration is not directly readable in channel number, a calibration chart shall be permanently affixed to the transmitter front panel. Filtered forced air cooling may be utilized for the power amplifier stages but air shall not pass internally through the cavities. The amplifiers shall be capable of operating into an open or shorted line without damage.

3.3.2.2.5 High level modulator. The high level modulator shall provide the keying sequence and drive to the tube stages to maintain proper output. Operator tuning adjustments shall not be required to maintain transmitter output performance criteria. Preventive or corrective maintenance adjustments shall be kept to a minimum and shall be made with the aid of a general purpose multimeter and oscilloscope only. A panel switch shall be provided to set the modulator for medium or high power output and the switch may be used to provide for single or dual stage operation. A second switch located internally may be required for the emergency operation mode of single stage operation.

3.3.2.2.6 Output power levels. The beacon shall be capable of operating (maintaining transmitter output characteristics of 3.3.2.2.7) at three power levels across the 962-MHz through 1213-MHz band as measured at the transmitter output connector. A front panel control may be provided to permit setting the power level to the tolerances specified:

Power	Source	Power output
Low	Solid-state amplifier	150 Watts (W) peak minimum
Mid	Intermediate power amplifier (IPA)	700 W peak ±10 percent
High	Power amplifier (PA)	3000 W peak ±10 percent

The IPAs amd PAs shall conform to their rated power outputs after 1000 hours of beacon operating time.

3.3.2.2.6.1 Emergency operation. The 150-W and 700-W operation shall be achieveable by disconnecting and reconnecting RF cables to bypass mid and high power stages. Additional cabling required for reconnecting shall be affixed to the inside of the transmitter drawer.

3.3.2.2.6.2 RF feed cable. The RF output cabling internal to the cabinet shall remain connected when the drawer is opened for servicing. Times Wire SF 142B cable and a retractor and slide assembly of the Jonathan 150 QDP button type, or equal, is recommended.

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3.3.2.2.6.3 Low pass filter. The common transmit and receive line shall contain a low pass filter to reject incoming and outgoing signals above 1600 MHz.

3.3.2.2.7 Transmitter output characteristics. Conformance to the transmitter output characteristics specified in 3.3.2.2.6 shall be achieved throughout the 962-MHz to 1213-MHz band at all three power levels.

3.3.2.2.7.1 Pulse shape. The pulse envelope, as detected by a linear detector, shall have a rounded shape falling within the limits of 3.3.2.2.7.1.1 through 3.3.2.2.7.1.4.

3.3.2.2.7.1.1 Pulse rise time. The pulse rise time is the time required for the leading edge of the pulse to rise from 10 percent to 90 percent of the maximum voltage amplitude. The pulse rise time shall nominally be 2.5 µs, but shall not exceed 3.0 µs. The minimum rise time is governed by the spectrum requirements of 3.3.2.2.7.2.

3.3.2.2.7.1.2 Pulse decay time. The pulse decay time is the time required for the trailing edge of the pulse to decay from 90 percent to 10 percent of the maximum voltage amplitude. The pulse decay time shall be nominally 2.5 µs, but shall not exceed 3.0 µs. The minimum decay time is governed by the spectrum requirements of 3.3.2.2.7.2.

3.3.2.2.7.1.3 Pulse top. The instantaneous amplitude of the pulse shall not, at any instant between the point of the leading edge which is 95 percent of the maximum voltage amplitude and the point of the trailing edge which is 95 percent of the maximum amplitude, fall below a line which is 95 percent of the maximum voltage amplitude of the pulse and is parallel to the baseline.

3.3.2.2.7.1.4 Pulse duration. The pulse duration measured between points on the leading edge and trailing edge of the pulse which are 50 percent of the maximum voltage amplitude of the pulse shall be 3.5 ±0.5 µs.

3.3.2.2.7.2 RF pulse spectrum. The energy level contained in each of two 0.5 MHz-bands centered on frequencies ±0.8 MHz above and below the channel frequency shall not exceed 200 milliwatts (mW), and the energy level contained in each of two 0.5-MHz bands centered on frequencies ±2.0 MHz above and below the channel frequency shall not exceed 2.0 mW.

3.3.2.2.7.3 Pulse droop. Prior to antenna amplitude modulation of the composite RF pulse train, the percentage modulation of the pulse train at 15 Hz and 135 Hz shall not exceed 1.0 percent for each. This modulation shall be measured with the transmitter operating normally and shall include the effects of droop and recovery time of the main and auxiliary reference groups. Spurious modulation at 120 Hz shall not exceed ±1.0 percent.

3.3.2.2.7.4 CW output. CW output between pulse pairs shall not exceed 5 microwatts with the beacon in normal operational mode. The peak signal amplitude for 1 µs between pulses of a pair on X-channels and 19 µs between pulses of a pair and pulses of the main reference group on Y-channel shall be at least 20 dB below the peak pulse amplitude. During the Y-channel auxiliary reference group, the energy shall be at least 20 dB below the peak pulse amplitude for at least 4 µs between pulses of the group.

3.3.2.2.7.5 Frequency stability. The transmitted frequency shall be maintained within ±0.002 percent of that specified in 3.3.2.2.1.

3.3.2.2.7.6 Duty cycle. The duty cycle shall conform to the requirements of continuous operation with the pulse repetition rate of 7200 pulses per s to 9800 pulses per s.

3.3.3 Beacon monitoring. Monitoring circuitry shall, whenever practical, be installed physically and electrically in close proximity to the beacon circuit being monitored. Monitoring circuitry shall not share the same cards as functional circuitry.

3.3.3.1 Monitoring alarms. Whenever a monitoring circuit detects an out of tolerance condition, an alarm signal shall be developed. This signal shall be programmable by switches or jumpers on the appropriate logic circuit card to deliver a hard or soft failure signal to the beacon control circuits. An internally adjusted delay from 5 s to 10 s shall activate the transponder control circuits. An internal switch shall be provided to inhibit this alarm. Placing the normal/test switch (see 3.3.5.3) in the test position shall also inhibit the alarm. An indicator displaying the monitor inhibit condition shall be provided on the transponder and on the control-indicator. Reply delay shall be wired as a hard failure. The alternating current (AC) line shall be wired as a hard failure, subject to the reset feature (see 3.3.6.1).

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3.3.3.2 Monitored parameters and tolerances. The parameters and tolerances shall be continuously and independently monitored as specified in TABLE II.

3.3.3.3 Monitor alarm indicators. All monitor alarm parameters shall be individually displayed on indicators that are vertically aligned on one side of cabinet panels and shall be prominently identified. Instantaneous failures shall be indicated by a 1-s memory and failures exceeding alarm limits shall be held in memory until manually reset. Alarm failure time limits shall be held in memory until manually reset.

3.3.3.4 Monitor fail-safe. A monitor fail-safe function shall be provided that exercises each of the monitored parameters to verify its error detection capability.

3.3.4 Beacon metering. Significant voltages, currents, and RF power levels shall be metered. The RF power metering shall be of a peak type and shall measure the low, medium, and high power levels at the system output with an accuracy of ±10 percent and through the range of 1000 pulse pairs per s to 5000 pulse pairs per s. General purpose test equipment (GPTE) may be used. Power supply voltage readings shall be centered on the meter scale within a green line tolerance area. Calibration adjustment shall be provided.

3.3.4.1 Metered transmitter levels. As a minimum, the levels specified in a through h shall be metered:

a. RF driver output power to an accuracy of ±1 dB
b. RF IPA output power to an accuracy of ±1 dB
c. RF PA output power
d. Reflected output power to an accuracy of ±1 dB
e. IPA and PA cathode current
f. IPA and PA cathode bias voltage
g. IPA and PA plate voltage
h. AC input line voltage

TABLE II. Monitored parameters and tolerances.

Parameter	Nominal value or range, or both	No alarm	Alarm
ID 1/ status			
Period	30 s	+10 percent	+100 percent
Continuous	0.100 s and 0.375 s	0.5 s	5.0 s
Pulse spacing	12 µs and 30 µs	±0.6 µs	±1.0 µs
RX sensitivity 2/ 3/	-90 dBm to -92 dBm	60 percent replies	50 percent replies
Duty cycle 3/	7200 pulses to 9800 pulses	3.3 percent	10.0 percent
Reply delay 4/	50 µs and 56 µs	±0.6 µs	±1.0 µs
AC line	60 Hz or 400 Hz, 115 V	±5 percent frequency 10 percent voltage	5/ 5/
Data link	45 s	<30 s percent	>60 s
Synthesizer status	5/		
Cavity temperature	5/ 6/		
Spectrum control	5/ (if feedback circuit used to control spectrum)		
Burst			
Main X	12 pulse pairs	7/	9/
Y	13 pulses	8/	10/
Aux X	6 pulse pairs	7/	9/
Y	13 pulses	8/	10/

Adjustment controls may be provided to set any of the tolerance limits; however, only a general purpose oscilloscope, counter, and multimeter shall be necessary.

1/ Identification
2/ Measured at -84 dBm
3/ The RX sensitivity and duty cycle alarm lights shall not flash when the ID tone is transmitted.
4/ Measured at -40 dBm
5/ Nominal values and tolerances to be determined by manufacturer; however, limits shall be within levels below those resulting in beacon out of tolerance conditions and resulting in damage to the equipment.

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6/ Cavity temperature shall be monitored by a temperature sensor. Monitoring of air flow is not required.
7/ Lose 1 pulse pair
8/ Lose 1 pulse
9/ Lose any 2 or more pulse pairs
10/ Lose any 2 or more pulses

3.3.4.2 Lamp test. A lamp test switch shall be provided for lamp testing. The requirements of MIL-STD-1472 apply.

3.3.4.3 Time totalizing meter. A time totalizing meter is required. This meter shall be in the tube filament line and shall display five significant digits in hours.

3.3.5 RF test generator. The monitor circuitry shall be capable of functioning as an RF test generator across the receiver band.

3.3.5.1 Synthesizer. The monitor synthesizer shall have the features specified in a through c:

a. Be directly interchangeable with the receiver-transmitter synthesizer
b. Have an externally programmable frequency control in MHz steps which is independent of the transmitter frequency control
c. Have a frequency deviation control to extend the interrogation band ±3 MHz in 100-kHz steps.

3.3.5.2 Monitor modulator. The monitor RF modulator shall be programmable to provide a variable pulse modulation from 60 pulse pairs per s to 4000 pulse pairs per s. A dial mark identifying the 2700-pulse pair rate on this range shall be provided. Fixed pulse pair spacings of 10 µs, 12 µs, 14 µs, 34 µs, 36 µs, and 38 µs, with a tolerance of ±0.5 µs, as well as a variable spacing control of from 10 µs to 40 µs, shall be provided.

3.3.5.3 Variable attenuator. A variable attenuator calibrated in 1-dB steps through the range from 0 dB to 100 dB, with an accuracy of ±0.5 dB, shall be provided. The variable attenuator shall be bypassed when not in the test position by a normal/test switch which shall also disable the programmable modulator pulse rate and spacing controls.

3.3.5.4 Test output. A panel-mounted test output jack shall provide a sample of the detected RF output when the normal/test switch is in either position.

3.3.6 Control circuits. The control circuitry shall provide the functions of transponder control, local and remote beacon system control, control-indicator, antenna group interface, and voice communication to the control-indicator (see 3.3.7b).

3.3.6.1 Transponder control. The transponder control circuitry shall provide for manual turn on, off, standby, and test operation, start up and shut down sequencing, automatic power failure, transient and overload protection, and monitor control. A test position shall connect the transmitter output to an internal dummy load. Indicators shall display beacon status of standby and on-air. When AC power at the equipment reaches the alarm tolerances of 3.3.3.2, the equipment shall automatically shut down before damage occurs. Whenever AC power is within tolerance, the equipment shall return to its previous ON state after approximately 10 s to 20 s (plus required tube warm-up) and the AC line monitor lamp shall illuminate as a soft failure indication. The high voltage arc detection circuits, when tripped, shall automatically reset three times and then remain open until manually reset. If operation is not interrupted for 20 s, the circuits shall automatically reset to the normal position. For power failures of short duration when the time of power loss is less than necessary to reinitiate power supply time delays, these delays shall be bypassed to achieve optimum on-air time without damage to equipment. The channel select switches shall be enabled only in the TUNE mode. Changing channels in the ON-AIR, TEST, or STANDBY modes shall enable the illegal channel alarm and automatically switch the beacon to the STANDBY mode.

3.3.6.2 Single and dual beacon system control. The system control circuitry shall be such that a dual beacon installation shall consist of two identical beacons requiring an additional control cable between the two to control priority of primary and secondary beacon operation. Operation shall be such that one beacon can be selected as the installation primary. Cable connections to the antenna, antenna control, status control indicator, and ancillary equipments shall be made from the selected beacon. The corresponding jacks on the secondary beacon shall be

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capped. Only the secondary beacon RF output (to the primary beacon) shall be a separate cable run. When the on-air beacon monitoring circuits sense a hard-failure, this beacon will transfer to standby with the RF output connected to its dummy load. The secondary beacon shall automatically be placed on the on-air condition. Automatic transfer shall be made in only one direction (from the operating beacon to the standby beacon). Only beacon function controls specific for that beacon shall exist. They may, however, change the status of the other beacon in the dual configuration. A suitable cover shall be provided to block unused indicators and switches for single beacon operation. Interconnection of major components shall be as specified in FIGURE 3 and FIGURE 4.

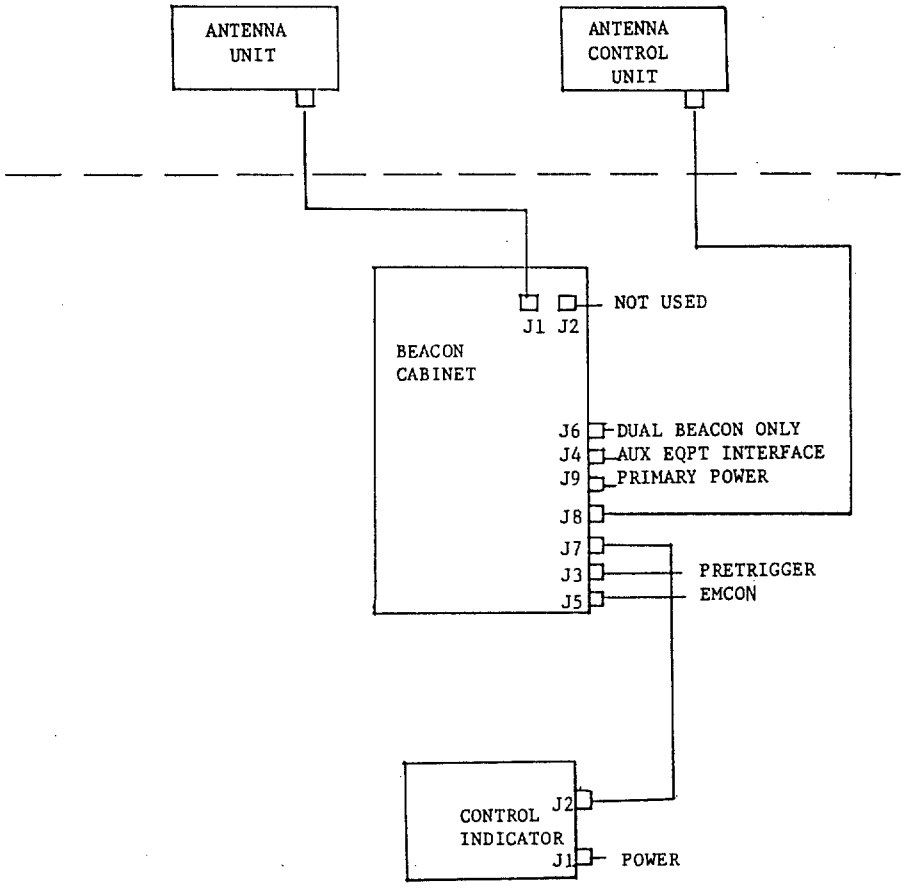


FIGURE 3. Single beacon interconnecting diagram.

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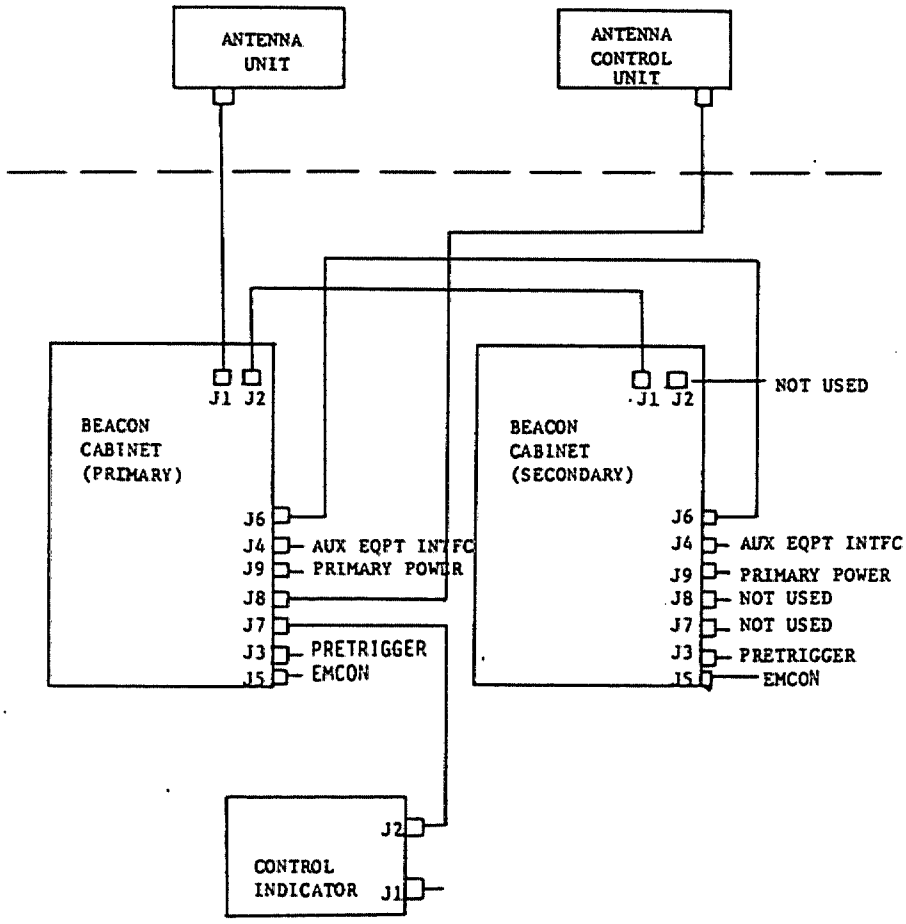


FIGURE 4. Dual beacon interconnecting diagram.

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3.3.6.2.1 Beacon and antenna group interface. The system of controls and interlocks, specified in a through f, shall be provided to interface the beacon with the antenna group in an operational installation:

a. The main and auxiliary reference triggers shall be provided to the beacon each by a separate shielded twisted pair (see 3.3.1.1.5).

b. A soft fail alarm signal shall be provided to the beacon by a contact open on a shielded twisted pair. Maximum load to be handled on this line is 28 VDC, 100 mA.

c. A hard fail alarm signal shall be provided to the beacon by a contact open on a shielded twisted pair. The maximum load to be handled on this line is 28 VDC, 100 mA.

d. A remote turn-on signal to the antenna group shall be provided by a +26 VDC level signal on a shielded twisted pair. Maximum load to be handled on this line is 200 mA at +28 VDC.

e. The beacon ready shall be provided to the antenna group on two separate shielded twisted pairs where logic 0 is -0.5 VDC to +0.5 VDC, 20 mA sinking current maximum. Logic 1 is +2.5 VDC to +5.0 VDC. Logic 1 shall be a minimum of +3 VDC with 1 mA source. A 01 signal shall represent the 3 kilowatt (kW) power, a 10 signal shall represent the 700-W power, a 11 signal shall represent the 150-W power, and a 00 signal shall represent beacon not ready.

f. Antenna band switching signals shall be provided to the antenna group on three separate shielded twisted pairs where logic 0 is -0.5 VDC to +0.5 VDC, 20 mA sinking current maximum. Logic 1 is +2.5 VDC to 5.0 VDC. Logic 1 shall be a minimum of +3 VDC with 1 mA source current. The switching logic shall be as specified in TABLE III.

TABLE III. Switching logic.

Frequency (MHz)	Y mode	Band code
962 to 993	0	00
994 to 1024	0	01
1025 to 1055	1	00
1056 to 1087	1	01
1088 to 1119	1	10
1120 to 1150	1	11
1151 to 1181	0	10
1182 to 1213	0	11

This signal shall be initiated by the synthesizer frequency setting switches. Maximum load to be handled on this line shall be +5 VDC, 100 mA.

3.3.6.2.1.1 Interface cable. The interface cable carrying the signals specified in 3.3.6.2.1 shall be a type 2SWU-12 with a maximum length of 15.24 m (50 ft). The mating connector to be supplied shall be a Bendix Part Number 10-214628-21S, or equivalent.

3.3.6.3 Emission control. The navigational set TACAN shall provide radiation status to a remote location, AN/SSQ-82(V) Mute system, through a separate connector on each transponder. This connector shall be in accordance with MS3114, Part Number MS3114E-12-10S with pin functions as follows:

Pin	Function
A	Deny command signal (+12 V) that silences the transponder until +12 V is removed
B	DC return for Deny command
C	Short circuit between C and D provides a closed circuit
D	When mating plug is connected
E	External voltage source (+12 V) for Pin F Deny command acknowledgment
F	+12 V from Pin E via closed relay contact acknowledging the receipt of a Deny command

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Pin	Function
G	Radiate indication signal from transponder to be +12 V from Pin H via relay contact closing whenever transponder is radiating
H	External source of +12 V for radiate indication signal
J	Ground from external equipment to be conducted to transponder ground

3.3.6.3.1 Functional operation. Functional operation shall be as specified in 3.3.6.3.1.1 through 3.3.6.3.1.3.

3.3.6.3.1.1 Monitor function. The monitor function requires that the transponder provide an indication of whether or not it is on the air; that is, any condition of output power: tune, test, low, mid, or high power. Closed relay contacts shall be provided between Terminal G and Terminal H. Contacts shall be rated for 50 VDC, 100 mA maximum.

3.3.6.3.1.2 Control function. The transponder shall be capable of having its output power capability being instantaneously silenced (Deny and Permit). The Deny command shall be actuated by the application of +12 V to Pin A with Pin B as the 12-V return. Removal of the +12 V to Pin A shall cycle the transponder from Deny to Permit returning the transponder to its normal operating status, with operator interaction permitted.

3.3.6.3.1.3 Acknowledge function. The transponder shall acknowledge the receipt of a Deny command by closing relay contacts between Pin E and Pin F and opening these contacts upon cycling back to Permit. Contact rating shall be 50 VDC, 100 mA maximum. Normally the external equipment will supply +12 V to Pin E.

3.3.7 Control-indicator. The control-indicator shall provide the indicators and controls to interface with the beacon as specified in a through g:

a. Provide control for one or two beacons to put either in standby or on-air and provide indicator lights for dark adapted areas to display status. An indicator to provide status of local and remote control shall be provided. Light sources shall be electrically dimmer controlled by a single adjustment. A lamp test switch shall be provided to test all lamps. A suitable cover shall be provided to block unused indicators and switches during single beacon operation. The standby beacon shall automatically be connected to a beacon internal dummy load. It shall be possible, in normal operation, to switch a beacon from on to standby and back to on in less than 5 s.

b. Provide two-way voice communication between the local and remote sites with an audio alarm operating in response to a ringing signal originating at either site. A telephone handset shall not be used. A panel mounted loud speaker and a nondetachable microphone shall be used. A Shure Model 488B on a retractable (coiled) cord with a push-to-talk (PTT) button, or equivalent microphone, is recommended. Activating only the PTT button shall allow a voice signal to substitute or multiplex with the status and control signals. A 45 s memory shall hold the status displayed on the indicators during voice if the voice is not multiplexed with the status signals. A loud speaker volume control shall be provided at both sites. Speaker power shall be a minimum of 2 W.

c. Provide monitor alarm signals, both aural and visual. A soft failure shall be indicated by a flashing amber light and a hard failure shall be indicated by a steady red light and, in addition, an aural alarm shall sound. An internal adjustment shall be provided that will allow a range adjustment to the aural alarm of at least 9 dB. An aural defeat switch shall be provided.

d. Provide control to put the on-air beacon in a medium or high power output mode; high power being the 3-kW level and medium power being the 700-W level.

e. Provide control and an indicator to provide operation of the Normal or Demand mode, or both.

f. Provide an indicator to display the monitor inhibit function which is controlled at the transponder only.

g. The control-indicator shall have the features specified in 1 through 3, as a minimum:

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1.	Alarms:	Hard fail indicator Soft fail indicator Transmission line fail indicator Aural alarm
2.	Status:	Monitor inhibit Remote control Emergency Mode (indicates 150 W or 700 W with single cavity) Service (for Demand Mode) On-air Standby
3.	Controls:	Normal or demand High or medium power On-air or standby Aural alarm defeat Lamp test Lamp dimmer Audio volume Ring initiate

3.3.7.1 Interface cable. The interface cable shall be a two-wire telephone line. Functions of the control-indicator shall be controlled over this line. For long runs this shall be a dedicated 600-ohm line with some lightning protection. Additional lightning protection within the equipment is desired. For distances up to 76.20 m (250 ft), a 120-ohm line may be used. An on-air status signal shall be available at J2 on the control-indicator. A closed contact shall indicate a beacon on-air condition. Relay contact rating shall be 1.0 ampere at 60 V.

3.3.7.2 Line signal characteristics. Line levels shall be adjustable to allow compatibility with world-wide commercial telephone line standards. Sufficient energy shall be available to allow transmission through wet-field wire with useful signals available at distances of at least 4.827 km (3 mi). The remoting system shall be fail-safe to the extent that any abnormality in the telephone line which would prevent its conveying the beacon status information to the remote location shall cause an alarm at the remote site. The operation of the beacon shall not be affected by the opening or shorting of the line. Loss of telephone line data, along with a power interrupt to the beacon shall cause the beacon to reinitialize in the STANDBY mode until telephone line data are resumed.

3.3.8 Ancillary equipment interface. The signals shall be available at an MS jack on the beacon cabinet cable entrance box as specified in a through f:

a. An identification keying signal of logic 0 is -0.5 VDC to +0.5 VDC, 20 mA sinking current maximum for the key up position and logic 1 is +2.5 VDC to +5.0 VDC, 20 mA source current maximum for the key down position.
b. A hard failure signal of logic 0 is -0.5 VDC to +0.5 VDC, 20 mA sinking current maximum for the failure alarm and logic 1 is +2.5 VDC to +5.0 VDC, 20 mA source current maximum for the no alarm condition.
c. A soft failure signal of logic 0 is -0.5 VDC to +0.5 VDC, 20 mA sinking current maximum for the failure alarm and logic 1 is +2.5 VDC to +5.0 VDC, 20 mA source current maximum for the no alarm condition.
d. A beacon ready signal of logic 0 is -0.5 VDC to +0.5 VDC, 20 mA sinking current maximum for ready condition and logic 1 is +2.5 VDC to +5.0 VDC, 20 mA source current maximum for the opposite condition.
e. An on-air signal of logic 0 is -0.5 VDC to +0.5 VDC, 20 mA sinking current maximum for a no hard or soft failure condition and logic 1 is +2.5 VDC to 5.0 VDC, 20 mA source current maximum for the failure condition.
f. Chassis ground

3.4 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with the Parts, materials and processes paragraph of MIL-E-16400, except that prior to selection and request for approval of nonstandard parts and materials, the procedures in MIL-STD-965, Procedure I shall be followed.

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3.4.1 Parts. Parts shall be as specified in 3.4.1.1 through 3.4.1.5.

3.4.1.1 Microelectronic devices. Microelectronic devices employed in the design of the navigational set requiring MIL-M-38510, Class B devices, shall conform to MIL-STD-454, Requirement 64.

3.4.1.2 Control locking devices. Control locking devices are required and shall be designed in accordance with the Shaft locking devices paragraph of MIL-E-16400, except that switches or other controls having detent mechanisms need not have locking devices.

3.4.1.3 Fuse locations. Fuses and spares for each shall be located on front panels.

3.4.1.4 Indicator switches. Flush-mounted indicator switches, if used, shall be serviceable from the front of the mounting panel.

3.4.1.5 Dial sizes and illumination. Dial sizes shall be proposed by the contractor and subject to the approval of panel layouts. Illumination of dials, switches, and controls shall be required only where necessary to permit the channel changing operation in darkened areas and shall be required for all functions of the control-indicator (see 3.3.7).

3.4.2 Processes. Processes shall be as specified in 3.4.2.1 and 3.4.2.2.

3.4.2.1 Painting. The exterior of the equipment shall be in accordance with the Painting paragraph of MIL-E-16400.

3.4.2.2 Soldering. Soldering shall be in accordance with WS-6536E with SCN 1.

3.5 Environmental conditions. The beacon shall operate properly within the service conditions for a sheltered controlled environment of MIL-E-16400, except as specified in a through h:

a. Ambient temperature (operating). -28° Celsius (C) to +65°C
b. Ambient temperature (nonoperating or storage). Range 4 of MIL-E-16400
c. Altitude. 0 m to 3048 m (10,000 ft) above mean sea level (MSL)
d. Altitude (nonoperating or storage). 0 m to 15,000 m (50,000 ft) above MSL
e. Salt fog (spray). The requirements of the Salt fog (spray) paragraph of MIL-E-16400 for sheltered equipment apply. In addition, samples of parts, components, and modules within the cabinet which would be exposed with the drawers open, shall be subject to the same requirements.
f. Humidity. MIL-STD-810, Method 507.2, Procedure 2
g. Shock. MIL-S-901, Grade A, Class I, Type A
h. Vibration. MIL-STD-167-1, Type I

3.6 Electrical characteristics. Electrical characteristics shall be as specified in 3.6.1 through 3.6.3.

3.6.1 Power source. The beacon shall operate within the requirements of this specification when connected to a source of power of either single-phase, 115 V, 60 Hz, or 115 V, 400 Hz having the characteristics as specified for Type I power of DoD-STD-1399, Section 300. There shall be no switches, cable changes, adjustments, or modifications permitted, except for primary power to main blowers. Primary input power circuits shall not be directly grounded and both sides of the line shall be fused.

3.6.2 Power supplies. The power supply outputs shall be protected by automatic cutouts during overvoltage and overcurrent. The power supplies shall not be damaged and shall return to operation upon removal of abnormal conditions. Power supplies shall be operated at no more than 80 percent of their rated values under full beacon load conditions. Fuses shall be used in lieu of circuit breakers on all power supplies. Combined fuse holders and fuse indicators shall be used. Loss of one or more supply voltages shall not result in damage to any single component, components, or higher assembly.

3.6.3 Interlocks. The interlocks shall not be cheatable with the beacon local-remote switch in remote position.

3.7 Mechanical characteristics. Mechanical characteristics shall be as specified in 3.7.1 through 3.7.1 8.

3.7.1 Equipment enclosures. Equipment enclosures shall be as specified in 3.7.1.1 through 3.7.1.6.

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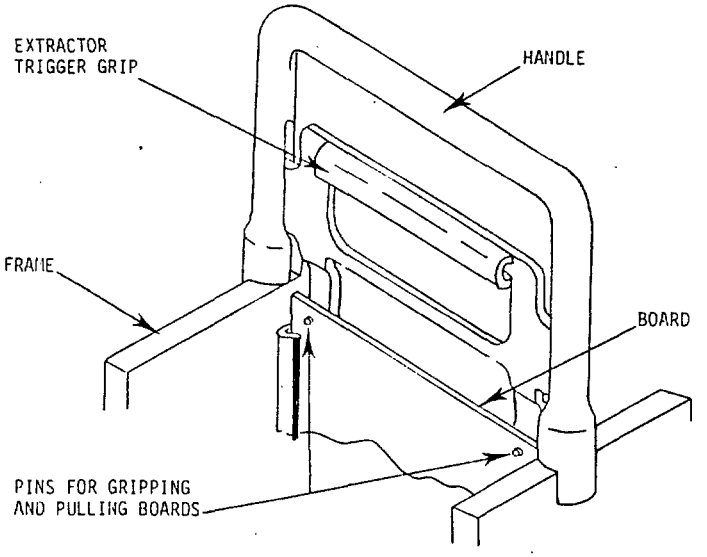
3.7.1.1 Equipment cabinet. The transponder cabinet shall have maximum dimensions of 1689.1 millimeters (mm) (66.5 inches (in.)) in height, (excluding removable hardware), 660.4 mm (26 in.) in depth, and 609.6 mm (24 in.) in width. The cable entrance and filter box shall be attached to the bottom of the right side of the cabinet as viewed from the front. The transponder cabinet weight shall not exceed 466 kilograms (1025 pounds). Shock blocks and guide pins shall be installed in the front and back of the equipment drawers.

3.7.1.1.1 Structural integrity. The structural integrity of the equipment, with base mounting only, shall be such that the self resonance shall be above 33 Hz in the vertical and front to back direction. The side to side axis shall be above 29 Hz.

3.7.1.2 Control-indicator. The control-indicator shall be contained in a single cabinet with a drawer suitable for reinstallation in a 482.6 mm (19 in.) rack in accordance with EIA RS-310-C-77.

3.7.1.3 Equipment fasteners. Each of the vertical equipment drawers shall have a single handle to facilitate quick opening and closing. The handle shall also provide a positive locking feature to secure the drawer in a closed position. Modules within the equipment drawers shall be secured by spring-loaded captive fasteners that resist loosening during vibration.

3.7.1.4 Maintenance access. Drawers that require maintenance actions on the underside shall have 90 degree rotatable slides with locking devices for the extended and rotated positions. Tilt slides shall be operable from the front panel. Doors and hinged chassis designed to swing out or down for maintenance access shall have braces and captive locking devices to secure the assembly in the maintenance position. Circuit cards shall be removable with a card puller of the type shown in FIGURE 5. Retaining hardware if used, shall be captive and finger-operable.



Recommended extractor tool equivalent to the types commercially available from Flotron Industries and Protolab (Positive Pin-Type pick up with spring loaded trigger).

FIGURE 5. Extractor tool.

3.7.1.5 Cable entrances. Cable entrances to the beacon cabinet shall be through MS connectors in accordance with MS3114 into the side-mounted cable entrance box. The RF input and output connectors shall be located on the top of the right side of the cabinet.

3.7.1.6 Degree of enclosure. The degree of enclosure shall be dripproof to an angle of 45 degrees.

3.7.1.7 Fans and blowers. If rotating devices are used for cooling, total protection against injury to maintenance personnel shall be provided.

3.7.1.8 Grounding. Grounding shall be in accordance with MIL-STD-1310, Class C, Type III straps.

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3.8 Airborne and structureborne noise. The requirement for airborne and structureborne noise shall be for Grade A, Type 3 equipment in accordance with MIL-STD-740.

3.9 Safety. Safety shall be in accordance with the Safety criteria paragraph of MIL-E-16400.

3.10 Leakage current. The leakage current of the navigational set shall not exceed 5 mA.

3.11 Electromagnetic compatibility (EMC) characteristics. The navigational set shall conform to the Class A4 requirements of MIL-STD-461, except that the REO2 transmitter frequency (f_0) level is relaxed by 35 dB. MIL-HDBK-241 may be used for guidance in electromagnetic interference reduction in power supply design.

3.12 Identification and marking. All panel marking shall be engraved.

3.13 Reliability requirements. The specified mean-time-between-failure (MTBF) (O_1 as defined by MIL-STD-781) shall be 1200 hours.

3.14 Maintainability. The navigational set shall have an equipment regain time (ERT) not exceeding 25 minutes when repair is accomplished at the organizational level of maintenance by replacement of faulty modules (including printed circuit boards (PCBs)) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, and checkout for all maintenance tasks. The ERT for piece-part repair of repairable modules at the intermediate or depot level of maintenance shall not exceed 120 minutes. The organizational level maintainability requirements shall be based on the capabilities (skill level and experience) of a Navy Electronics Technician, third class (ET-3) in accordance with NAVPERS 18068.

3.14.1 Maintenance concept. The maintenance concept for the navigational set at the user level is for repair by replacement at the module level utilizing the built-in-test and indication features, the module and chassis test points, and GPTE. Factory only adjustments shall not exist.

3.14.2 Test measurement and diagnostic equipment (TMDE) (see 6.4.6). Test provisions for the equipment shall conform to MIL-STD-454, Requirement 32, except as specified herein.

3.14.2.1 Test point criteria. Except for ship testing, test points shall be accessible with the removal of not more than one cover panel while the equipment is operational and without removal of cables or connectors. The use of extender boards is permitted to allow access to test points on printed circuit cards. Test points shall be provided for the requirements specified in a through e. (Multiple use of test points is permitted and unintentional grounding of test points shall not cause damage to the equipment.)

a. Signal quality and circuit performance. Test points, test jacks, or both, shall be provided to permit the injection of signals and the monitoring of signals at the input/output (I/O) terminals of the system (see 6.4.5), each unit, and each assembly (see 6.4.3).

b. Alignment. Test points, test jacks, or both shall be provided to energize, inject a standard dynamic signal, or monitor the intermediate circuit affected by the alignment. Where TMDE is used to align or adjust a system, unit, assembly, and subassembly (see 6.4.2) circuitry shall be designed so that a technician may see and operate the TMDE while making adjustments and probing the circuit under test.

c. Fault isolation. Test points, test jacks, or both, shall be provided to permit the injection and measurement of signals at I/O terminals of the unit, assembly, and subassembly to determine satisfactory operation or malfunction of the unit, assembly, or subassembly.

d. Shop testing. Shop testing of assemblies and subassemblies shall be performed with the assemblies and subassemblies removed from the prime equipment. Test points shall be provided so that the assembly may be activated (such as direct current (DC) and dynamic signals applied), checked for quality of operation, and fault isolated to the shop replaceable item (that is, flatpack, voltage controlled oscillator, mixer, detector, multiplier, and so forth). Assembly and subassembly test points shall be provided in a single connector. Assembly and subassembly test points may be placed at various nodes or locations on the assembly and subassembly.

e. Standard test jacks. The standard test jacks specified in 1 through 4 shall be provided as a minimum:

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1. Detected RF output (composite signal)
2. Sync (main trigger)
3. Receiver output (video)
4. Coder output (video)

3.14.2.2 General purpose electronic test equipment (GPETE). GPETE for use at all levels of maintenance shall be selected from the standard and substitute standard GPETE listed in MIL-STD-1364. The procedures of MIL-STD-1387 shall apply when the GPETE listed in MIL-STD-1364 will not produce a satisfactory test.

3.14.2.3 Special tools. The need for special tools (see 6.4.7) shall be kept to a minimum. If required, special tools shall be supplied with each equipment, shall be in accordance with MIL-STD-454, Requirement 63, and shall conform to the storage and operating environmental conditions specified for the prime equipment.

3.15 Workmanship. Workmanship shall be as specified in 3.15.1 and 3.15.2.

3.15.1 General workmanship. Workmanship shall be as specified in MIL-STD-454, Requirement 9.

3.15.2 Workmanship screen. Navigational sets shall withstand a defect detection vibration screen of random vibration of $0.02g^2$/Hz ±3 dB from 350 Hz to 2000 Hz, with a roll-off of 3 dB per octave starting at 350 Hz, conducted as specified in 4.8.2, and temperature cycling as specified in 4.8.1 and 4.8.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through d:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Safety verification (see 4.5.2)
d. Inspection of preparation for delivery (see 4.10)

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4.3 First article inspection. Unless otherwise specified (see 6.2), one single and one dual navigational set shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine conformance with the requirements of this specification. First article inspection shall include the examination and testing specified in TABLE IV and the performance tests specified in TABLE V, Groups A, B, and C. The total test requirements for first article inspection may be proportioned over the contract's first articles, subject to approval of the procuring activity.

TABLE IV. Examination and test.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination						
Weight	3.7.1.1	4.5	X			
Dimensions	3.7.1.1	4.5	X			
Parts	3.4.1	4.5	X			
Painting	3.4.2.1	4.5	X			
Identification and marking	3.12	4.5	X			
Safety	3.9		X			
General workmanship	3.15.1		X	X		
Performance	TABLE V		X	X	X	X
Temperature sequence						
Low temperature	3.5	4.5	X			X
High temperature	3.5	4.5	X			X
Humidity	3.5	4.5	X			X
Salt fog (spray)	3.5	4.5	X			X
Shock	3.5	4.5	X			
Vibration	3.5	4.5	X			
Enclosure	3.7.1.6	4.5	X		X	
Steady-state voltage and frequency	3.6.1	4.5	X		X	
Transient voltage	3.6.1	4.5	X		X	
Transient frequency	3.6.1	4.5	X		X	
Spike voltage	3.6.1	4.5	X		X	
Power interruption	3.6.1	4.5	X		X	
Power and power factor	3.6.1	4.5	X			
Leakage current	3.10	4.5.3	X			X
EMC	3.11	4.5.4	X			X
Maintainability	3.14	4.7 1/	X			
Reliability	3.13	4.6	X			
Thermal survey		4.5.1 1/	X			
Airborne and structureborne noise	3.8	4.5	X			

1/ Required on original development contract only.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed as specified in 4.4.1 through 4.4.3.3.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each navigational set offered for delivery. The inspection shall comprise examinations and tests which will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of material. Production inspection shall include the examinations and tests of TABLE IV, Group A, and the performance tests specified in TABLE V, Group A.

4.4.2 Production control inspection (Group B). Production control inspection shall be performed on a randomly selected sample from each month's production lot and shall include examinations and tests of TABLE IV, Group B, and the performance tests specified in TABLE V, Group B. Production control inspection shall be performed on navigational sets that have passed production inspection. The sample shall satisfactorily complete the production control inspection prior to release of the month's production to shipment.

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TABLE V. Performance tests.

Tests	Requirement paragraph	Quality conformance inspection		
		Group A	Group B	Group C
Peak power out	3.3.2.2.6	X		
RF pulse spectrum	3.3.2.2.7.2		X	
Pulse shape	3.3.2.2.7.1	X		
Pulse droop	3.3.2.2.7.3		X	
Frequency stability	3.3.2.2.7.5	X		
CW output	3.3.2.2.7.4		X	
Pulse coding	3.3.1	X		
Squitter rate	3.3.2.1.10	X		
Demand tolerances	3.3.2.1.3		X	
Squitter distribution	3.3.2.1.14			X
Receiver bandwidth	3.3.2.1.4			X
Echo suppression	3.3.2.1.9		X	
Receiver selectivity	3.3.2.1.5			X
Single pulse rejection	3.3.2.1.6			
Narrow pulse interference	3.3.2.1.12			X
CW interference	3.3.2.1.11		X	
Response delay	3.3.2.1.7	X		
Triggering level	3.3.2.1.2	X		
Dead time	3.3.2.1.8		X	
Receiver decoding	3.3.2.1.6		X	
Monitor tolerances	3.3.3.2	X		
BITE 1/ tolerances	3.3.5	X		
Controls and interlocks (beacon)	3.3.6	X		
Controls and interfaces	3.3.7	X		
Interface (antenna)	3.3.2.2.1.3, 3.3.6.2.1	X		
Pretrigger tolerances	3.3.1.1.6		X	
Ancillary interface	3.3.8	X		

1/ Built-in-test equipment (see 6.4.1)

4.4.3 Environmental inspection (Group C). Environmental inspection shall be as specified in TABLE IV, Group C, and the performance tests specified in TABLE V, Group C and shall be performed on units which have been subjected to and have passed production inspection. Vibration testing from 33 Hz to 50 Hz shall be limited to a single sweep search for resonance at a reduced magnitude.

4.4.3.1 Sampling for environmental inspection. The initial sample shall be selected from the first month's production lot, and shall satisfactorily complete environmental inspection prior to release to shipment of the second month's production lot. Subsequently, one each sample shall be selected from each successive 25 navigational sets, or fraction thereof, and shall pass environmental inspection prior to release to shipment of succeeding month's production quantities.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and details of the corrective action to be taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. The contractor shall demonstrate the effectiveness of the corrective action to the QAR by subjecting two additional navigational sets to the failed environmental inspection. In the case of the humidity test, only the two additional corrected cards and modules or components shall be subjected to the same number of hours and

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conditions as for the system test. If the contractor cannot demonstrate effective corrective action that enables the equipment to conform to specified requirements, or if the cause of failure cannot be determined, the matter shall be referred to the contracting officer.

4.4.3.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract (see 6.2), sample units which have been subjected to, and have passed, environmental tests may be accepted on the contract, provided they are resubjected to, and pass, production inspection after repair of all damage.

4.5 Test methods. Examinations and tests specified in TABLE IV and TABLE V shall be conducted to verify conformance to applicable specification requirements and as specified in 4.5.1 through 4.5.4.

4.5.1 Thermal survey. To verify the system design portion of 3.6a, the contractor shall conduct a complete thermal survey of the navigational set during temperature environmental testing, to determine the existence of hot spots which may prove to be injurious to the useful life of the system. The survey shall be conducted with the system packaged and enclosed in the proposed final production configuration. The survey shall include a thermal profile of the internal configuration of the system and all critical items at both 25°C ambient and the maximum specified ambient operating temperature. The survey shall be completed and all required corrective measures taken by the contractor prior to approval by the procuring activity. A hot spot is any item (part or assembly of parts) used in the system including, but not limited to, semiconductor devices, printed circuit boards (PCBs), transformers, and resistors which attain a temperature in excess of that recommended by the item manufacturer or applicable specification, or an assembly of parts that attain a temperature which could cause accelerated degradation of the system. The thermal survey shall be repeated whenever system changes occur which may affect this characteristic.

4.5.2 Safety verification. Examinations, analyses, and tests shall be performed to verify conformance to 3.9.

4.5.3 Leakage current test. The leakage current test shall be performed as specified in 4.5.3.1 and 4.5.3.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE NAVIGATIONAL SET DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRIC SHOCK PROTECTION.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.3.1 Navigational set connections. Each navigational set directly connected to an external power source and units deriving power from the navigational set shall be placed on an insulated surface. All safety ground conductors between the navigational set and units deriving power from the navigational set shall be intact. The safety ground conductor between the navigational set and the source power shall be opened during the test. The navigational set shall be connected as shown in FIGURE 6 if it is connected to single-phase source power.

4.5.3.2 Measurement. Leakage current shall be measured on the navigational set in its normal operating configuration. Navigational set controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the navigational set is designed. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in FIGURE 6. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.4 EMC test. Conformance to the requirements of 3.11 shall be verified by tests performed in accordance with MIL-STD-462.

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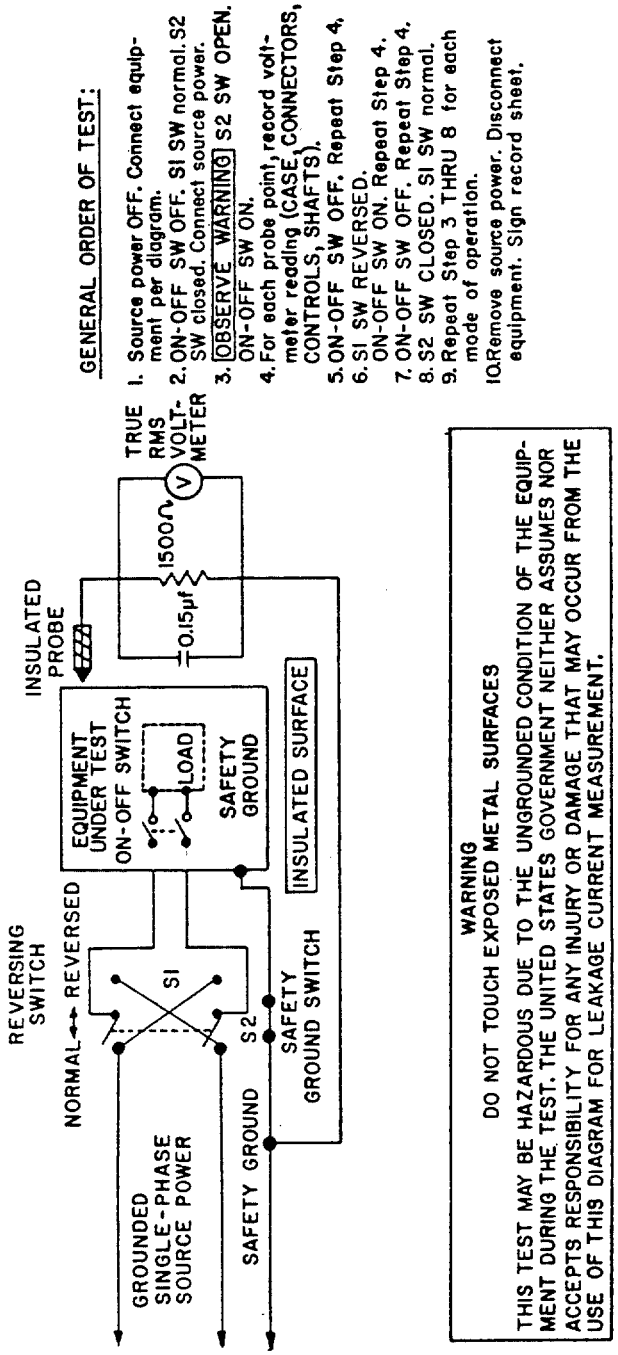


FIGURE 6. Single-phase test diagram for leakage current measurement.

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4.6 Reliability qualification test (RQT). An RQT shall be conducted in accordance with the general requirements of MIL-STD-781 and Test Plan XXIC. Testing shall proceed to a decision in accordance with accept or reject criteria of Test Plan XXIC.

4.6.1 RQT conditions. The navigational sets shall be subjected to the environmental conditions of MIL-STD-781, equipment Category 3A and the Test conditions matrix table of MIL-STD-781.

4.6.2 Rejection. If the test results in a reject decision, the procuring activity shall be immediately notified. The contractor shall, at no cost to the Government,

a) rerun the entire RQT, or
b) rerun portions of the RQT to verify the effectiveness of the corrective action developed to remedy causes for the rejection. Successful demonstration of corrective actions without additional failures shall be a basis for hardware acceptance.

4.6.3 Failure. The final decision as to the definition and classification of failures shall be made by the procuring activity, using the Failure categories paragraph of MIL-STD-781 as a guide. Any operating discrepancy or degradation that requires an unscheduled adjustment or calibration to be made after initial satisfactory operation of the navigational set shall be defined as a failure.

4.7 Maintainability demonstration. Conformance with the equipment and module maintainability requirements shall be demonstrated in accordance with MIL-STD-471, Test Method 4. Only Phase II of MIL-STD-471 shall apply.

4.7.1 Equipment maintainability. The equipment maintainability demonstration shall consist of 20 corrective maintenance tasks requiring replacement of modules (or PCBs) or chassis-mounted parts at the organizational level of maintenance. Fifty faults shall be determined for the equipment maintainability demonstration in accordance with APPENDIX A of MIL-STD-471. The procuring activity or its representative shall select 20 of these faults to be used in this part of the maintainability demonstration. For fault simulation, a number shall be assigned to each component and part within a module or chassis-mounted component and part. Fault simulation for modules shall be accomplished by introduction to faulty parts corresponding to the number selected or by simulating such faults. As part of the demonstration, the accessibility shall be observed to determine conformance with 3.14.

4.7.2 Module maintainability. The module maintainability demonstration shall consist of 20 corrective maintenance tasks requiring repair of modules (or PCBs) by replacement of parts at the intermediate or depot level of maintenance, as appropriate. A minimum of two faults shall be determined for each repairable module. Fault selection and simulation for each repairable module shall be in accordance with the procedure for the equipment maintainability demonstration. Using the list as a guide, a total of 20 faults shall be selected by the procuring activity for the module maintainability demonstration.

4.7.3 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the equipment maintainability demonstration. The module maintainability demonstration shall be performed by contractor personnel.

4.7.4 Technical documentation. Technical documentation shall be limited to the technical manual.

4.7.5 Rejection. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies, and
b. Upon approval of an approach to the redesign or corrections, the contractor shall repeat the demonstration until an accept decision is reached.

4.8 Workmanship. Workmanship shall be as specified in 4.8.1 through 4.8.4.

4.8.1 General workmanship. The navigational set shall be examined to determine conformance to the requirements of 3.15.1.

4.8.2 Workmanship screen. Workmanship screen shall consist of the three segments specified in a through c:

a. Nonoperating temperature cycling at the subassembly level. All modules shall be subjected to nonoperating temperature cycling.
b. Random vibration at the system level. One navigational set from each third month's production shall be subjected to random vibration screening.

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c. Temperature cycling at the system level. Temperature cycling at the system level shall be conducted on each system.

4.8.2.1 Nonoperating temperature cycling at the subassembly level. Subassemblies shall be categorized as nonhybrid and hybrid as specified in 4.8.2.1.1 and 4.8.2.1.2. High voltage power supply potted modules shall not be subjected to nonoperating temperature cycling.

4.8.2.1.1 Nonhybrid facility assemblies. Nonhybrid facility assemblies shall be as specified in a through f:

a. Two chambers shall be used to conduct thermal shock tests. One chamber shall be set at -54°C, hereinafter referred to as cold chamber, and one chamber shall be set at +100°C, hereinafter referred to as hot chamber.
b. Thermal shock shall be conducted chamber to chamber, with minimum dwell time between chambers.
c. The unit under test shall be placed in the cold chamber and shall remain for 1 hour.
d. After 1 hour in the cold chamber, the unit under test shall be removed and placed in the hot chamber and remain for 1 hour.
e. Procedures c and d, run sequentially as specified herein, complete one cycle of thermal shock.
f. Twenty-five cycles shall be conducted on each unit under test.

4.8.2.1.2 Hybrid facility assemblies. Hybrid facility assemblies shall be as specified in a through c:

a. Two chambers shall be used to conduct thermal shock tests. The cold chamber shall be set at -28°C and the hot chamber shall be set at +90°C.
b. Procedures b through d, run sequentially as specified in 4.8.2.1.1, complete one cycle of thermal shock.
c. Fifteen cycles shall be conducted on each unit under test.

4.8.3 Vibration. Following the subassembly nonoperating temperature cycling, and prior to conducting temperature cycling at the operating cabinet level, vibration shall be performed. The vibration shall be random, or, subject to procuring activity approval, pseudo-random or complex waveform vibration for an accumulated time of 10 minutes in the top to bottom direction on the OX-52/URN and in the front to back direction on the C-10363/URN. All test items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions specified herein. The control accelerometer shall be located next to one of the mounting points of the item under test. Navigational sets having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density. The test item shall be energized during vibration. Failures occurring during vibration shall be corrected and the test resumed. If random vibration testing is conducted at a facility remote to the contractor's plant, only failures related to loss of operating voltages shall be corrected on site. Any remaining failures may be corrected after return to the contractor's plant. Failures occurring in the sample navigational sets shall be localized and identified so that all other like navigational sets produced that month may be examined or tested for the presence of like defects.

4.8.4 Temperature cycling at the system level. Following vibration, if applicable, each navigational set shall be subjected to the temperature cycle shown in FIGURE 7. The temperature rate of change shall be not less than 5°C per minute. Navigational set power shall be turned on and off at the indicated times. When practical, navigational set drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. Cooling blowers within the navigational set may also be operated when the navigational set is turned off. When a test is called for, a functional operating test shall be performed. The dwell time for each cycle shall be 80 percent of the time required for the largest electrical or electronic part to become temperature-stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition. When test failures occur, the navigational set shall be reworked and the cycling continued for a cumulative total of six cycles.

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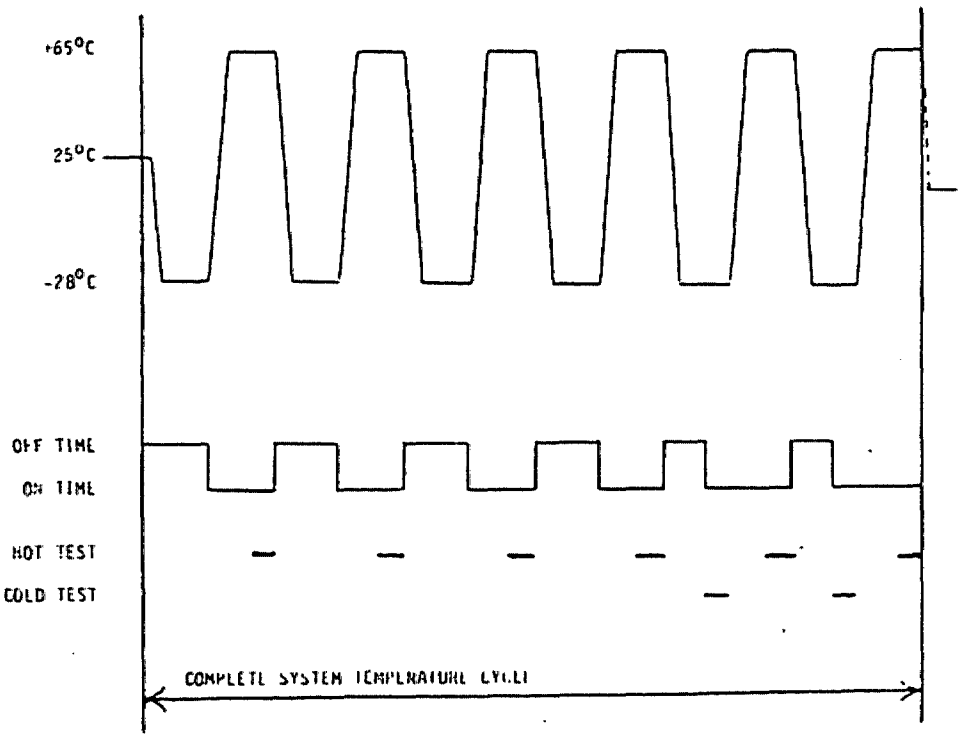


FIGURE 7. Temperature cycle at the system level.

4.9 Manufacturing run-in. Following workmanship screen, each navigational set shall be subjected to a manufacturing run-in. This test shall consist of at least 40 operating hours at ambient temperature with the last 24 hours failure-free.

4.10 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The navigational set, covered by this specification, is intended to be interfaced with a family of TACAN antennas for service on ship or shore facilities. The navigational set will provide distance, azimuth, and identification information to aircraft with properly equipped interrogators.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. When reinspected environmental sample units may not be accepted (see 4.4.3.3)
d. Applicable levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of one single and one dual navigational set. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. Definitions of terms used in this specification are given in 6.4.1 through 6.4.8.

6.4.1 BITE. Instruments, devices (electrical, electronic, mechanical, magnetic, or chemical, and so forth), tools and interface equipment, built as an integral part of the equipment and used to: test, observe, measure, monitor, modify, generate, calibrate and control physical properties, movements, and other characteristics. This includes power sources, modifiers, scalers, jigs, templates, monitoring devices, plumbing standards, and programming software required to test, measure, and diagnose.

6.4.2 Subassembly. Two or more parts which form a portion of an assembly, but having parts which are individually replaceable (such as, modules and PCBs with components).

6.4.3 Assembly. A number of parts or subassemblies or any combination thereof, joined together to perform a specific function and capable of disassembly (such as, audio oscillator, amplifier, synthesizer, crystal oscillator).

6.4.4 Unit. An assembly or any combination of parts, subassemblies, and assemblies mounted together, normally capable of independent operation in a variety of situations. For the purpose of this specification, the following items are units: Each base station equipment, that is, interface central unit, system control unit, RF repeater, remote monitor unit; also portable equipment including protective communications helmet, transceiver electronics, F/P transceiver, F/V charger amplifier, damage control adapter, and support equipment including the cryptovariable fill unit, and battery charger.

6.4.5 System. All possible combinations of units required to perform a function.

6.4.6 TMDE. TMDE are all instruments, devices (electrical, electronic, mechanical, magnetic, or chemical, and so forth), tools and interface equipment used to: test, observe, measure, monitor, modify, generate, calibrate and control physical properties, movements, and other characteristics. This includes power sources, modifiers, scalers, jigs, templates, monitoring devices, waveguides, standards, and programming software required to test, measure, and diagnose.

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6.4.7 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

6.5 Refurbishing first article units. After first article approval, first article units should be refurbished and should pass the production inspection tests.

6.6 Subject term (keyword) listing.

Antenna tuning triggers
AN/URN-25, navigational set
Auxiliary reference pulse group
Distribution statement A
Electromagnetic compatibility (EMC)
General purpose electronic test equipment (GPETE)
Leakage current
Main reference pulse group
Maintainability
Navigational set, AN/URN-25
Noise airborne and structureborne
Pretrigger
Pretrigger characteristics
Pulse coding
Reliability
Safety
TACAN
Test measurement and diagnostic equipment (TMDE)
Workmanship screen

6.7 Supersession data. This specification includes the requirements of ELEX-T-238B, dated 14 June 1983.

Preparing activity:
NAVY - EC

(Project No. 5825-N020(EC))

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